Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)		No net force (1) No net moment (1) Accept forces must add up to 0 / all forces must be balanced / all moments must be balanced / ACM = CM Don't accept forces must be equal or moments must be equal	2			2		
	(b)		$(2 \times 8) - 11.8$ (1) Weight of beam = 4.2 [N] (1) Or: Correct moment equation (1) Correct answer (1)		2		2		
	(c)		**Method 1**: Taking moments about B $F_A \times 1.8$ (1) $(7.2 \times 1.4) + (4.2 \times 0.9) + (11.8 \times 0.15)$ (1) $F_A = 8.7$ N (1) $F_B = (4.2 + 11.8 + 7.2)$ (1) – 8.7 (**ecf**) **OR** FB $\times 1.8 = (7.2 \times 0.4) + (4.2 \times 0.9) + (11.8 \times 1.65)$ (1) $F_B = 14.5$ N (1) **ecf** **Method 2:** As method 1, but taking moments about A N.B. If all correct except F_A and F_B the wrong way round award 4 marks **Method 3:** Taking moments about centre (W): Correct clockwise expression e.g $0.9F_A + 8.85$ (1) Correct anticlockwise expression e.g. $0.9F_B + 3.6$ (1) Correct sum of forces expression: e.g. $F_A + F_B = 23.2$ (1) Correct algebra to show $F_B = 14.5$ N (1), $F_A = 8.7$ N (1)		5		5	5	
			Question 1 total	**2**	**7**	**0**	**9**	**5**	**0**